

Day 5: Power Query Data Cleaning Techniques

📌 Definition to Remember

Data Cleaning in Power Query entails resolving missing values (`null`), handling data errors, deduplicating records, splitting compound strings, and creating rule-based conditional columns.

1. Core Data Cleaning Techniques

1. Handling Missing Values (`null`):

- *Fill Down / Fill Up*: Replaces nulls in grouped hierarchy tables with preceding values.
- *Replace Values*: Explicitly replaces `null` with `0` for metrics or `"Unknown"` for text categories.

2. Error Handling:

- Identifying data type mismatch errors (`[Error]`).
- Using *Remove Errors* vs *Replace Errors* with default fallback values.

3. Removing Duplicates:

- Deduplicating on composite business keys (e.g., `CustomerID + OrderDate`) to guarantee dimension uniqueness.

4. Splitting & Merging Columns:

- *Split by Delimiter*: Splitting `"First Last"` by space into separate First Name and Last Name columns.
- *Merge Columns*: Combining Address, City, and Postal Code into a standardized Full Address.

5. Conditional Columns:

- Visual GUI rule builder creating `IF-THEN-ELSE` classification flags (e.g., `IF Age >= 60 THEN "Senior" ELSE "Adult"`).

★ Must-Master Skills & Interview Points

1. What causes Power Query data type errors (e.g., text characters in numeric columns) and how to handle them.
2. Difference between removing duplicate rows across the whole table vs specific key columns.
3. Best practice: Never load raw uncleaned data into the modeling tab.

⚡ Quick Recall

Deduplicate → Fill Down Nulls → Handle Errors → Split/Merge Strings → Add Conditional Columns